

## SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS  
COURSE CODE: CSA07

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### COURSE OUTCOME COVERED

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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### Annexure A

### Debugging & Optimisation Task

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#### Code of Conduct:

I V. Sanjay (192312486) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

V. Sanjay  
Signature of the student:

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### Annexure A

### Debugging & Optimisation Task



1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

#### Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

#### Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

#### Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.

- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

#### Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

| Destination Network | Next Hop |
|---------------------|----------|
| 192.168.1.0/24      | 10.0.0.2 |
| 192.168.2.0/24      | 10.0.0.3 |
| Default Route       | 10.0.0.1 |

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

#### Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

#### RUBRICS FOR CALCULATION

| Criteria | Weightage | Level 4<br>(Exemplary) | Level 3<br>(Proficient) | Level 2<br>(Developing) | Level 1<br>(Beginning) |
|----------|-----------|------------------------|-------------------------|-------------------------|------------------------|
|          |           |                        |                         |                         |                        |

|                        |     |                                                                                      |                                                        |                                               |                                               |
|------------------------|-----|--------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|
| Problem Identification | 20% | Accurately identifies all errors and root causes with clear understanding            | Identifies most errors with minor gaps                 | Identifies some errors but misses key issues  | Unable to identify errors correctly           |
| Debugging              | 25% | Applies systematic debugging techniques effectively to resolve issues                | Uses appropriate debugging methods with minor errors   | Limited debugging approach; requires guidance | Unable to debug or resolve issues             |
| Optimization           | 20% | Optimizes code by significantly improving time complexity using efficient algorithms | Improves time complexity with minor gaps in efficiency | Limited improvement in time complexity        | No improvement; inefficient approach retained |
| Analysis               | 20% | Demonstrates strong logical reasoning and clear justification of solutions           | Shows reasonable analysis with some justification      | Limited analysis; weak reasoning              | No analytical reasoning demonstrated          |
| Code Quality           | 15% | Produces clean, well-structured code with proper comments and documentation          | Code is mostly clear with minor documentation gaps     | Code lacks structure and proper documentation | Poorly written code with no documentation     |

## 4. Network Configuration Debugging.

IP address = 192.168.10.65

Subnet Mask = 255.255.255.192 (128)

Default Gateway = 192.168.10.1

### 1. Identify the subnet:

⇒ /26 subnet size = 64 addresses

192.168.10.0/26 (0-63)

192.168.10.64/26 (64-127)

192.168.10.128/26

192.168.10.192/26

### 2. Network Details.

Network Address: 192.168.10.64

Broadcast Address: 192.168.10.127

Valid Host Range: 192.168.10.65 -

192.168.10.126.

### 3. Verify IP and Gateway:

⇒ IP 192.168.10.65 → Valid.

⇒ Gateway 192.168.10.1 → Invalid.

### 4. Configuration Error.

⇒ IP address = 192.168.10.65

⇒ Default Gateway = 192.168.10.128.

### 5. Optimized Configuration:

⇒ IP address = 192.168.10.65

⇒ Default Gateway = 192.168.10.128.

6. Usable Hosts,

total addresses = 64

Network + Broadcast = 2.

Usable Hosts = 62.

② IPv6 Configuration Debugging

Device A = 2001:db8:ff00:42::8329/64

" B = 2001:db8:0:0:1::1/64.

1. Expanded Address.

Device A:

2001:0db8:0000:0000:0000:ff00:0000:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

2. Network Prefix:

Device A → 2001:db8:0:0::/64

" B → 2001:db8:0:0::/64

3. Same Subnet?

both belong to 2001:db8:0:0::/64

4. Addressing Error

communication failure is likely due to incorrect gateway, firewall, or Neighbor Discovery issues.

5. Optimized Configuration,

Device A → 2001:db8:0:0::1/64

" B → 2001:db8:0:0::2/64

6. Available Hosts:

Host Bits = 64.

Available addresses = 264

= 1844674407370955166



⑤ Subnet Allocation Optimization.

192.168.1.0/24.

| Subnet | Network      | Host Range.                     |
|--------|--------------|---------------------------------|
| 128    | 192.168.1.0  | 192.168.1.1 -<br>192.168.1.63   |
| 124    | 192.168.1.64 | 192.168.1.65 -<br>192.168.1.94  |
| 128    | 192.168.1.96 | 192.168.1.97 -<br>192.168.1.110 |

Unused addresses:

- 128 → 64 total (62 usable)
- 124 → 32 total (30 usable)
- 128 → 16 total (14 usable)

Inefficient Allocation.

→ Small department using 28 waste many addresses.

→ Large department using 128 may not have enough hosts.

Suboptimizing:

Increased subnet assignment causes address shortage.

Optimized VLSM:

Design subnets according to department req.  
→ large department → /26 (65 hosts)  
→ Medium " → /27 (32 hosts)  
→ small " → /28 (16 hosts)

④ IPv6 Configuration Debugging (Same as Q2).

Expanded Address:

Device A:

2001:db8:0000:0000::0000:1100:2004:2329

Device B:

2001:db8:0000:0000:0001:0000:0000:0001

Network Prefix:

2001:db8:0:0::/64

Same Subnet:

Yes.

Optimized Address:

Device A = 2001:db8:0:0::0/64

Device B = 2001:db8:0:0::1/64

Available Hosts:

2<sup>64</sup> = 18446744073709551616

## Routing Table Debugging:

Destination IP:

192.168.2.100

Solution:

1. Longest Prefix Match  
Matches 192.168.2.0/24.

2. Next Hop

10.0.0.3.

3. Verify Destination.

192.168.2.0 /24.

4. Possible Routing Error.

Next-hop router (10.0.0.3) may be

unreachable or incorrectly.

B. Optimized Routing:

Verify connectivity to 10.0.0.3.

Ensure route 192.168.2.0/24 → 10.0.0.3 is

correct.

Remove any duplicate or incorrect  
routes.

6. If No Working Route Exist.

Router was the Default Route →

10.0.0.1